Draft appendix: the model of the m_{hh} spectrum and its uncertainty

1 November 2026 (v4: smooth model)

A The model of the m_{hh} spectrum and its uncertainty

The di-Higgs cross section as a function of m_{hh} and of the HEFT parameters $\theta = (\kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$ can be written as [1]

$$\frac{d\sigma}{dm_{hh}}(m_{hh};\theta) = \sum_{i=1}^{23} C_i(\theta)\, a_i(m_{hh}), \tag{1}$$

where the C_i are known monomials in θ and the 23 coefficient functions a_i depend on m_{hh} only (section **??**). In this appendix we describe how the a_i are determined from POWHEG spectra, how the uncertainty of the resulting model is obtained, and how both are validated on POWHEG spectra not used in the fit.

A.1 Data and noise

The data are 3,989 POWHEG spectra at 13 TeV for the fit and 1,000 independent spectra for the test, each at a different point θ, in 80 bins of 15 GeV between $m_{hh} = 255$ and 1455 GeV. Each spectrum is a histogram of 40,000 generated events of equal weight q_s (a few incomplete runs have fewer), so the cross section in bin k is exactly $t_{sk} = q_s n_{sk}$ with an integer count n_{sk}. We recover q_s and the counts of every spectrum as the greatest common divisor of its bin contents: all spectra are integer multiples of a single q_s to the precision of the stored numbers. The counts are Poisson distributed and, up to a negligible multinomial correlation within a spectrum, independent. Near threshold the bins hold about 1,000–1,800 events, at 1.4 TeV about 20. Bins with no events are included in the fit.

Of the training file, 88% of the rows (281,044 including 1,814 empty bins) are used in the fit; the remaining 38,076 rows, never used in the fit, serve to determine the model discrepancy (appendix A.4).

A.2 The model

Each coefficient function is expanded in $J = 40$ cubic B-splines [2],

$$a_i(m) = \phi(m) \sum_{j=1}^{J} \beta_{ij} B_j(m), \qquad \phi(m) = \sqrt{1 - 4m_h^2/m^2}, \tag{2}$$

with breakpoints spaced uniformly in $\sqrt{m - 2m_h}$, i.e. denser just above threshold. The factor ϕ, the velocity of the Higgs bosons in their rest frame, makes the density vanish at threshold as the two-body phase space does. The model's prediction for a bin is the *integral* of eq. (1) over the bin,

$$f_{sk} = \sum_{i=1}^{23} \sum_{j=1}^{J} C_i(\theta_s)\, \beta_{ij}\, I_{jk}, \qquad I_{jk} = \int_{\text{bin } k} \phi(m) B_j(m)\, dm, \tag{3}$$

not its value at the bin centre, so the model is a genuine density in m_{hh}. The prediction is *linear* in the $23J = 920$ parameters β_{ij}. With one box-shaped basis function per bin in place of the splines, eq. (3) becomes a free fit of 23 numbers per bin; this "free fit" sets the floor that any model of the form (1) can reach, and we use it as a benchmark.

A.3 Fit and posterior

Likelihood. With hundreds of events in most bins a Gaussian likelihood with the Poisson variance is accurate, but the variance must be taken from the model, $q_s \max(f_{sk}, q_s)$, not from the observed count: weighting each bin by $1/(q_s n_{sk})$ (Neyman's χ^2) favours downward fluctuations and biases the fit low by about one event per bin [3], which with 3,989 spectra pooled is many standard deviations at high mass. We therefore iterate: fit, recompute the variances from the fit, refit (iteratively reweighted least squares [4]). At convergence this is the Poisson maximum-likelihood fit.

Smoothness prior. For each function a Gaussian prior on the second differences of its coefficients [5], $-\ln\pi(\boldsymbol{\beta}_i) = \frac{1}{2}\lambda_i \sum_j (\beta_{ij} - 2\beta_{i,j-1} + \beta_{i,j-2})^2$ plus a negligible ridge term, the discrete analogue of a penalty on $\int (a_i'')^2\, dm$. The 23 strengths λ_i are chosen by maximising the evidence [6], which is available in closed form because the model is linear.

Posterior. Likelihood and prior being Gaussian in $\boldsymbol{\beta}$, the posterior is *exactly* Gaussian, with mean $\hat{\boldsymbol{\beta}} = H^{-1}\Phi^{\mathsf{T}} W \boldsymbol{t}$ and covariance H^{-1}, $H = \Phi^{\mathsf{T}} W \Phi + \Lambda$. Because the 23 monomials and the 80 bins factorise, $\Phi^{\mathsf{T}} W \Phi = \sum_k S_k \otimes \boldsymbol{I}_k \boldsymbol{I}_k^{\mathsf{T}}$ with $S_k = \sum_s w_{sk} \boldsymbol{C}_s \boldsymbol{C}_s^{\mathsf{T}}$, and the whole fit takes about a minute on a laptop. The data determine about 600 of the 920 combinations of parameters (the number of well-determined parameters [6]). Predictions for any binning, the density at any m_{hh}, the coefficient functions and the correlations between bins follow from $\hat{\boldsymbol{\beta}}$ and H^{-1}. The fit uncertainty of a bin is small, a median 2.9% of the Poisson noise. The results do not change, within 0.01 in any measure below, for $J = 30$ or 60. The model should be used within 250–1455 GeV: above the fitted range the splines are fixed by the prior alone (appendix A.5).

A.4 Model discrepancy

On data not used in the fit the residuals exceed the noise and the fit uncertainty, most of all for spectra whose cross section is a small difference of large terms. Ranking the test spectra by the ratio $\sigma_{\mathrm{tot}} / \sum_k \sum_i |C_i a_{ik}|$, the most cancelling fifth has $\langle z^2 \rangle = 1.70$ against 1.01 for the least cancelling, and its residuals are correlated between neighbouring bins. The same is true of the free fit, so the excess is not a property of the model but a departure of the POWHEG spectra from the form (1), consistent with a small relative inaccuracy, smooth in m_{hh}, of each of the 23 components. We therefore add a discrepancy proportional to the size of the individual terms,

$$s^2_{\mathrm{disc},k} = \varepsilon(m_{hh})^2 \Big(\sum_{i=1}^{23} |C_i(\theta)\, a_{ik}| \Big)^2, \tag{4}$$

in the spirit of Kennedy and O'Hagan [7], with ε determined bin by bin on the 38,076 rows not used in the fit by requiring the pulls to have unit variance, and $\ln\varepsilon$ smoothed by a quadratic in m_{hh}. It rises from 0.08% at threshold to 0.28% at 600 GeV and 3.7% at 1.45 TeV. A discrepancy proportional to the noise instead, $a(m_{hh})\sigma_k^2$, gives the same overall calibration but is too narrow for the cancelling spectra and too wide for the others. The discrepancy is treated as independent between bins; for predictions summed over many bins its correlation (over about 5–8 bins near threshold) must be taken into account.

	$-2\ln\lambda$ per bin		median
	255–555 GeV	>555 GeV	per spectrum
smooth model	1.566	1.100	1.078
free fit per bin (benchmark)	1.559	1.105	1.075
network ensemble (cross-check)	1.561	1.121	1.085

Table 1: Accuracy on the 1,000 test spectra: the mean likelihood-ratio statistic $-2\ln\lambda$ per bin (1 for a perfect model with Poisson noise), and its median per spectrum. The network was trained with observed-count weights (appendix A.3), which costs it above 555 GeV.

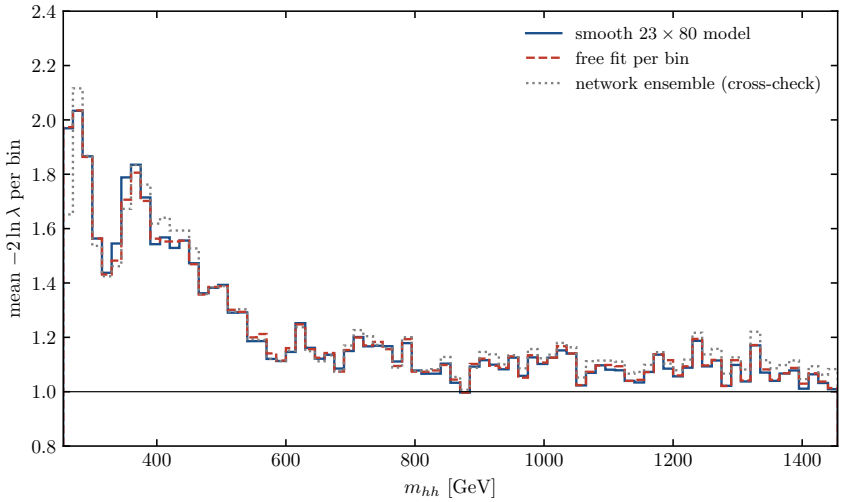


Figure 1: Mean likelihood-ratio statistic $-2\ln\lambda$ per bin on the test spectra as a function of m_{hh}; 1 for a perfect model with Poisson noise.

A.5 Validation on the test spectra

Accuracy. We measure the accuracy of the prediction in each bin by the likelihood-ratio statistic

$$-2\ln\lambda = -2\ln\frac{\text{Poisson}(n;\mu)}{\text{Poisson}(n;n)} = 2\left[\mu - n + n\ln\frac{n}{\mu}\right], \tag{5}$$

where $n = t/q$ is the POWHEG event count and $\mu = f/q$ the model's prediction. If POWHEG were exactly the model plus Poisson noise, $-2\ln\lambda$ would follow a χ^2 distribution with one degree of freedom (Wilks' theorem), with mean 1; no parameter is adjusted to the test spectra. (Statisticians call this quantity the Poisson deviance.) The smooth model reaches the floor set by the free fit: on the 1,000 test spectra the mean of $-2\ln\lambda$ per bin is 1.566 below and 1.100 above 555 GeV (table 1, figure 1). The excess over 1 is the departure discussed in appendix A.4; it is largest just above threshold and for the spectra with the strongest cancellation. Figure 5 shows examples.

Calibration. With the discrepancy the pulls have unit standard deviation within 1%, overall, below and above 555 GeV, and at every degree of cancellation (table 2, figure 2). The total predictive uncertainty is a median 1.014 times the Poisson noise of a POWHEG spectrum; it is dominated by the discrepancy where the terms cancel and by nothing at all elsewhere.

	s.d. of pull				$\langle z^2 \rangle$ by cancellation fifth	
	all	255–555	$>$ 555 GeV	$\lvert z \rvert < 2$	strongest	weakest
noise + fit uncertainty	1.090	1.218	1.043	93.9%	1.70	1.01
+ discrepancy, eq. (4)	1.004	1.011	1.001	95.4%	0.97	1.00

Table 2: Calibration on the test spectra: $z = (t - f)/\sqrt{qf + s^2 + s^2_{\text{disc}}}$ with s the exact fit uncertainty. For a standard normal, $|z| < 2$ holds for 95.4%. With the discrepancy, $\langle z^2 \rangle$ is 0.97, 0.98, 1.04, 1.04, 1.00 in the five fifths.

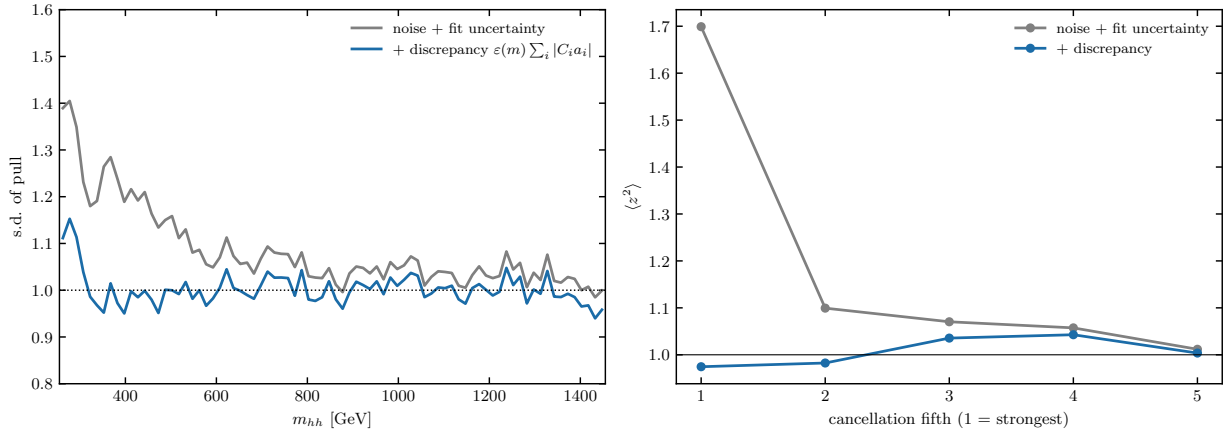


Figure 2: Standard deviation of the pull as a function of m_{hh} (left) and $\langle z^2 \rangle$ by degree of cancellation (right), without and with the discrepancy.

Inside a bin. To test the model as a density, we fitted it to the training spectra merged into 30-GeV bins and predicted the 15-GeV bins of the test spectra. Above about 360 GeV it predicts how each 30-GeV bin splits into its two halves as well as the model fitted to 15-GeV bins ($\chi^2 \approx 1$ per pair), where a free fit, which can only halve each 30-GeV value, is off by up to a factor of 30 in χ^2 (figure 3). Below 345 GeV the threshold rise has structure on scales below 30 GeV, which 30-GeV data cannot resolve.

The edges. The POWHEG files also contain the bins 240–255 and 1455–1500 GeV, not used in the fit. Summed over the test spectra, the model predicts the first to within 8% (without the factor ϕ, 55% too high); the second, an extrapolation fixed by the prior, is predicted 5–44% too low depending on J, within the correspondingly large uncertainty.

The coefficient functions (figure 4) are smooth, defined at every m_{hh}, and agree with the network for all but the least constrained ones. The functions whose monomials are small or nearly degenerate over the sampled θ, a_{16}–a_{20}, are determined to 20–75%; the smoothness prior improves the worst of them (a_{16}, a_{17}) by factors of about 1.3 and 3.6 relative to the network.

A.6 Cross-checks

Network ensemble. A neural network for the 23 functions, with the same structure (1), trained with an L1 loss and fine-tuned with the Huber loss [8] of the pulls, as an ensemble of 20 networks [9],

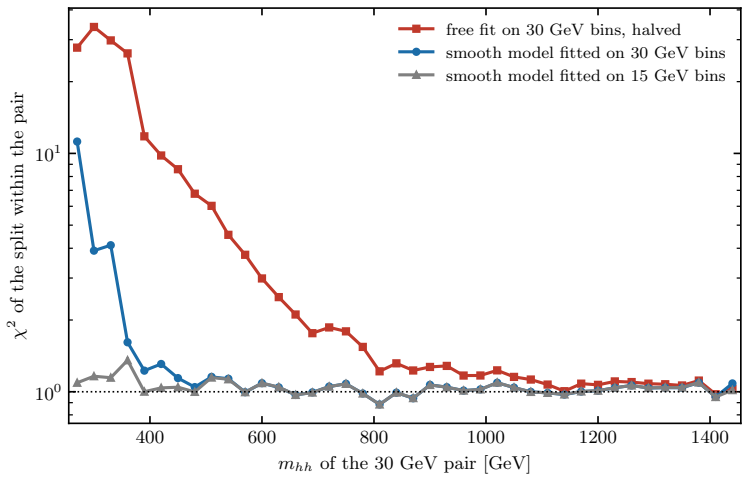


Figure 3: Within-bin accuracy. The training spectra were merged into 30-GeV bins and the 15-GeV bins of the test spectra predicted. Shown is the χ^2 of the split of each 30-GeV pair into its two halves (1 is perfect).

reaches the same accuracy below 555 GeV (table 1). Its spread is 5.6% of the noise, 80% of it due to the random initialisation and 20% to the finite training sample (from a comparison of bootstrap [10] and seeds ensembles). With the discrepancy (4) its pulls are calibrated as well (s.d. 1.004). Between the bin centres, where no data constrain it, the networks disagree twice as much as at the centres below 400 GeV; the smooth model does not have this problem.

Bayesian network. Hamiltonian Monte Carlo [11, 12] over the network parameters [13] gave a posterior mean of the same accuracy but a width of only 0.5% of the noise, because a single chain could not explore a posterior whose curvatures span many orders of magnitude. A mixture of linearised Laplace approximations [14, 15] about the 20 networks gives 2.4% within and 5.6% between networks, comparable to the fit uncertainty of the smooth model, and, like it, needs the discrepancy to be calibrated.

A.7 Summary

The cross-section model used in this paper is the smooth 23×80 model: the exact HEFT form (1) with each coefficient function a smoothed spline times the threshold factor, fitted to the bin integrals of 3,989 POWHEG spectra by Poisson maximum likelihood with smoothness chosen by the evidence. Its posterior is exactly Gaussian. On 1,000 independent spectra it is as accurate as any model of this form can be, and with a model discrepancy proportional to the size of the individual terms, determined on data not used in the fit, its uncertainty is calibrated to 1% at all m_{hh} and at every degree of cancellation. Its fit uncertainty is a few per cent of the POWHEG noise; the remaining uncertainty reflects the consistency of the POWHEG spectra with the 23-term form, not the model.

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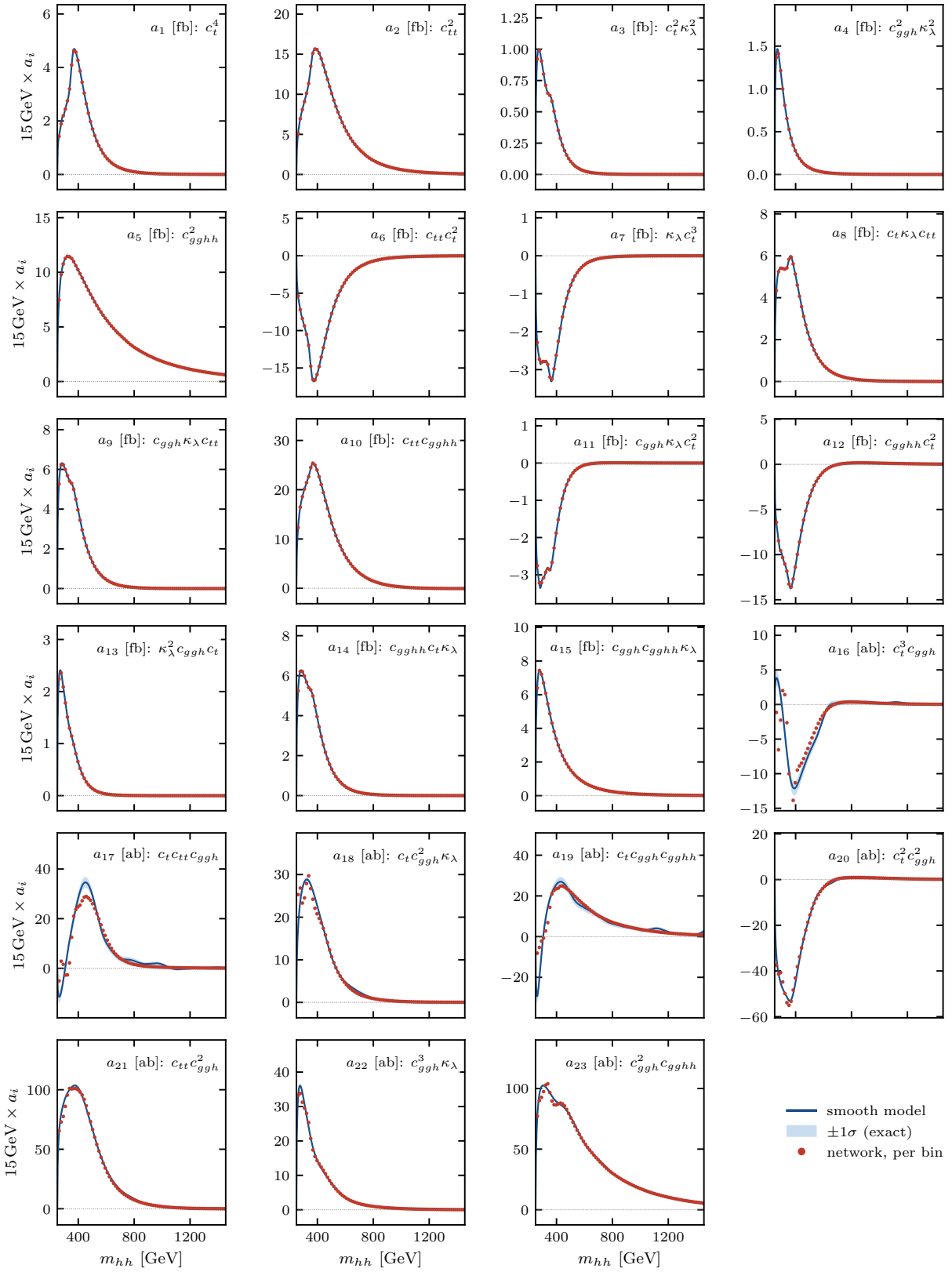


Figure 4: The 23 coefficient functions $a_i(m_{hh})$, shown as $15\,\mathrm{GeV} \times a_i$ (the cross section per 15 GeV multiplying the monomial in each panel), with their exact $\pm1\sigma$ bands, and the per-bin values of the network ensemble (red points).

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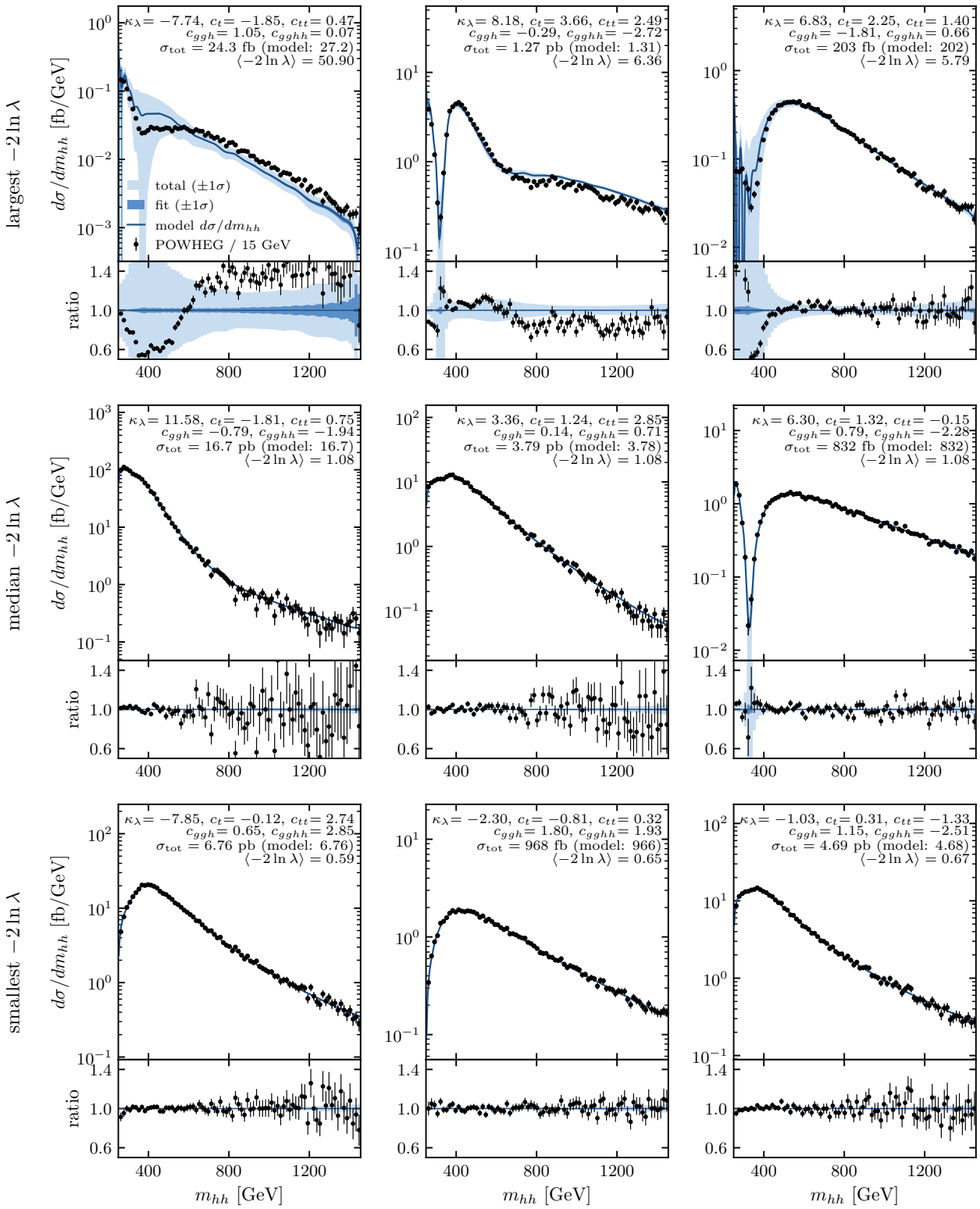


Figure 5: POWHEG spectra divided by the bin width (points; error bars: Poisson noise) and the smooth model's density $d\sigma/dm_{hh}$ (line) for nine of the 1,000 test spectra, ranked by the mean likelihood-ratio statistic $-2\ln\lambda$ per bin (equation 5): the three largest (top), three around the median (middle) and the three smallest (bottom). Dark band: fit uncertainty ($\pm1\sigma$); light band: total, including the model discrepancy. The lower panels show the ratio of POWHEG to the model's bin integrals. Each panel gives the HEFT parameters, the total cross section in 255–1455 GeV from POWHEG and from the model, and the mean of $-2\ln\lambda$ per bin. The largest values occur where the total cross section is smallest, through destructive interference.